

Competitive Programming

Section 42: Advanced Data Structures Mixed Contest

Competitive Programming Course

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1 Advanced structure diagnosis

These tasks are about choosing what information must remain available after many operations. The right structure follows from the query contract.

2 Complete local contest set

Problem	Title
A	Dynamic Order Statistics
B	Range Add Range Sum
C	Static Rectangle Count
D	Nested Ranges Count
E	Subtree Add, Point Query
F	Dynamic Path Maximum
G	Subtree Value Count
H	Connectivity Countdown
I	Persistent Version Sums
J	Subtree K-th Smallest
K	Sparse Rectangle Sums
L	Path Add, Path Maximum

Problems A–F establish the core coverage. Problems G–L deliberately increase in difficulty, moving from static range indexing to reversed time, persistence, sparse two-dimensional indexing, and lazy heavy-light decomposition.

3 Model the operations first

For each local problem, list every update and query, then answer:

```
Is the input static or dynamic?  
Does a query ask for an aggregate, a position, or a count of objects?  
Can coordinates be compressed without changing comparisons?  
Can queries be processed offline?  
What information must an internal node or prefix preserve?
```

Do not turn this into a lookup table from wording to structure. Derive an invariant that would let a proposed representation answer every operation. The editorial contains the local construction after you have tested yours.

4 Prove and test the invariant

State the meaning of each stored number in one sentence. Show how an update preserves it, and why the query combines stored values into exactly the desired answer. Test against a slow vector or nested-loop oracle on tiny random cases, especially repeated values, empty ranges, and coordinates near boundaries.

5 Cost accounting

Give both per-operation and total complexity. Include preprocessing, compression, sorting, and memory. A logarithmic query can still be unsuitable if a hidden rebuild happens for every operation.

6 Exercises

For A through L, write an operation contract and an invariant before coding. Also name one boundary case your brute-force oracle must include. Consult the editorial only after this attempt. Solve G–L in order: the progression is part of the exercise design.